Sergey Lisenkov

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Education

2005 **Ph.D. Physics**, Institute of Biochemical Physics, Russian Academy of Sciences, Moscow, Russia

2001 **M.S. Physics**, Volgograd State University, Volgograd, Russia

Appointments

2013– **University of South Florida, Tampa**
Assistant Research Professor, Department of Physics

2008–13 **University of South Florida, Tampa**
Visiting Research Assistant Professor, Department of Physics

2005–08 **University of Arkansas, Fayetteville**
Postdoctoral Researcher, Department of Physics

Research Areas

- Prediction, design and optimization of properties of semiconductors and ferroelectric materials
- Using state-of-the-art first-principles methods to study electronic, structural, optical, dielectric and piezoelectric properties
- Developing new tools for large-scale simulations

Grants and Awards

2022–26 Collaborative Research: U.S.–Ireland R&D Partnership "Antiferroelectricity, Ferrielectricity and Ferroelectricity in the Archetypal Antiferroelectric $PbZrO_3$ at Small Scale", National Science Foundation, DMR-2219477 ($255,000, PI)

2020–23 "Unveiling intrinsic functionality of two-dimensional organic-inorganic ferroelectrics for energy storing/converting devices: integrated computational-experimental approach", National Science Foundation, ECCS-2029800 ($450,000, co-PI)

2010–20 Faculty International Travel Grant, University of South Florida ($2,500 every year)

2009 New Research Grant, University of South Florida ($6,000, PI)

Courses Taught

Graduate

- Statistical Mechanics (Spring 2024)
- Statistical Mechanics II (Fall 2024)

Undergraduate

- Statistical Mechanics (Spring 2020, 2021, 2025, 2026)
- Mathematical Methods in Physics (Spring 2017; Fall 2017, 2022, 2023)
- Classical Mechanics (Fall 2018; Spring 2019; Fall 2019, 2020)
- General Physics (Spring 2018)
- Enriched General Physics (Spring 2016, Fall 2016)

SERVICE

Academic Journal Peer Review

Physical Review Letters, Physical Review B, Physical Review Materials, Applied Physics Letters, Nano Letters, Journal of Chemical Physics

Service to the Field

2019 Co-organizer, international workshop *Fundamental Physics of Ferroelectrics and Related Materials*

Service to the University

2018– USF Physics Graduate Admission Committee

PUBLICATIONS

Preprints

2026 Arlies Valdespino, Abduljelili Popoola, **Sergey Lisenkov**, and Inna Ponomareva, "Negative and Zero Linear Compressibility in MCN (M = Ag, Au, Cu): A First-Principles Study", arXiv:2607.06904.

Journal Articles and Conference Proceedings

94 peer-reviewed publications, including 13 in *Physical Review Letters*, 27 in *Physical Review B*, 6 in *Physical Review Materials* and 3 in *Nano Letters*, and papers in *Nature*, *Nature Materials*, *Nature Communications*, *JACS* and *Advanced Materials*. 5,043 citations, h-index 33 (Google Scholar, October 2026).

2026

[94] Nikhilesh Maity, **Sergey Lisenkov**, Arlies Valdespino, Milan Haddad, Lewys Jones, Amit Kumar, Nazanin Bassiri-Gharb, and Inna Ponomareva, "Relaxation approach to quantum-mechanical modeling of ferroelectric and antiferroelectric phase transitions", *Nature Communications* **17**, 10883 (2026).

[93] Min-Chul Kang, Joshua Townsend, **Sergey Lisenkov**, Inna Ponomareva, Xiaoli Tan, and Lin Zhou, "Direct Observation of Structural Disorder Near Critical Point in $BaTiO_3$ Crystal", *Physical Review Letters* **137**, 016102 (2026).

[92] Ravi Kashikar, Joshua Townsend, **S. Lisenkov**, and I. Ponomareva, "DFT-based insight into finite-temperature properties of ferroelectric perovskites with lone-pair: the case of $CsGeX_3$ (X = Cl, Br, I)", *Journal of Physics: Condensed Matter* **38**, 015403 (2026).

2025

[91] Abduljelili Popoola, Ravi Kashikar, Ali Azmy, Ioannis Spanopoulos, Homayoun Jafari, Jagoda Sławińska, Sarath Witanachchi, **Sergey Lisenkov**, and Inna Ponomareva, "Mechanically and electrically tunable Rashba-Edelstein effect in ferroelectric semiconductors, $CsGeX_3$ (X = I, Br, Cl)", *Physical Review Materials* **9**, 084412 (2025).

[90] Salma Khatun, Nikhilesh Maity, **Sergey Lisenkov**, Inna Ponomareva, and Matthias Batzill, "Alloying of Mn with Bi-Rich Surfaces of Bi_2Te_3 by Topotactic Reaction", *Small* **21**, e04498 (2025).

[89] Maggie Kingsland, J. Townsend, Sina Najmaei, **Sergey Lisenkov**, and Inna Ponomareva, "First-principles effective Hamiltonian for finite-temperature modeling of nonperovskite ferroelectrics", *Physical Review B* **112**, 064104 (2025).

[88] Milan Haddad, Vasily Lebedev, Kristina Holsgrove, Sergio Rivera-Cruz, Sarah Stock, Nikhilesh Maity, **Sergey Lisenkov**, Inna Ponomareva, Amit Kumar, Lewys Jones, and Nazanin Bassiri-Gharb, "Chemical Compensation Challenges in Processing Antiferroelectric $PbZrO_3$ Thin Films", *ACS Omega* **10**, 29878–29887 (2025).

[87] Rinku Paul, Nikhilesh Maity, Biswajit Das, Seema Rani, Kaushik Ghosh, **S. Lisenkov**, Inna Ponomareva, and Ranajit Ghosh, "Controllable oxygen vacancy defect engineering of $BiVO_4$ porous structures for room temperature NH_3 detection", *Chemical Engineering Journal* **515**, 163814 (2025).

[86] Nikhilesh Maity, Milan Haddad, Nazanin Bassiri-Gharb, Amit Kumar, Lewys Jones, **Sergey Lisenkov**, and Inna Ponomareva, "Ferroelectricity at the extreme thickness limit in the archetypal antiferroelectric $PbZrO_3$", *npj Computational Materials* **11**, 48 (2025).

2024

[85] A. Popoola, N. Maity, R. Kashikar, **S. Lisenkov**, and I. Ponomareva, "Large Electrically and Chemically Tunable Rashba-Dresselhaus Effects in Ferroelectric $CsGeX_3$ (X = Cl, Br, I) Perovskites", *Journal of Physical Chemistry C* **128**, 17806 (2024).

[84] Ravi Kashikar, Arlies Valdespino, Charlton Ogg, Edvin Uppgard, **S. Lisenkov**, and I. Ponomareva, "Ferroelectricity in Ultrathin Halide Perovskites", *Nano Letters* **24**, 10624 (2024).

[83] Nikhilesh Maity, Ravi Kashikar, **S. Lisenkov**, and I. Ponomareva, "Giant spin splitting and its origin in methylhydrazinium lead halide perovskites", *Physical Review Materials* **8**, L091402 (2024).

[82] Ravi Kashikar, D. DeTellem, P. S. Ghosh, Yixuan Xu, S. Ma, S. Witanachchi, Manh-Huong Phan, **S. Lisenkov**, and I. Ponomareva, "Coupling of structure and magnetism to spin splitting in hybrid organic-inorganic perovskites", *Journal of the American Chemical Society* **146**, 13105 (2024).

[81] Joshua Townsend, Ravi Kashikar, **S. Lisenkov**, and I. Ponomareva, "Ferroelectric phases and phase transitions in $CsGeBr_3$ induced by mechanical load", *Physical Review B* **109**, 094121 (2024).

[80] Ravi Kashikar, **S. Lisenkov**, and I. Ponomareva, "Coexistence of polar and antipolar phases in ferroelectric halide perovskite $CsGeBr_3$", *Physical Review B* **109**, L020101 (2024).

[79] M. Kingsland, **S. Lisenkov**, S. Najmaei, and I. Ponomareva, "Phase transitions in HfO_2 probed by first-principles computations", *Journal of Applied Physics* **135**, 054102 (2024).

2023

[78] Vimukthi Pathirage, Salma Khatun, **Sergey Lisenkov**, Kinga Lasek, Jingfeng Li, Sadhu Kolekar, Manuel Valvidares, Pierluigi Gargiani, Yan Xin, Inna Ponomareva, and Matthias Batzill, "2D Materials by Design: Intercalation of Cr or Mn between two VSe_2 van der Waals Layers", *Nano Letters* **23**, 9579 (2023).

[77] Ravi Kashikar, Abduljelili Popoola, **S. Lisenkov**, A. Stroppa, and I. Ponomareva, "Persistent spin textures in halide perovskites induced by uniaxial stress", *Journal of Physical Chemistry Letters* **14**, 8541 (2023).

[76] Abduljelili Popoola, Partha Sarathi Ghosh, M. Kingsland, Ravi Kashikar, D. DeTellem, Yixuan Xu, S. Ma, S. Witanachchi, **S. Lisenkov**, and I. Ponomareva, "First-Principles Property Assessment of Hybrid Formate Perovskites", *Journal of Chemical Physics* **159**, 074702 (2023).

[75] Yulian Yao, Aaron Naden, Mengkun Tian, **Sergey Lisenkov**, Zachary Beller, Amit Kumar, Josh Kacher, Inna Ponomareva, and Nazanin Bassiri-Gharb, "Ferrielectricity in the Archetypal Antiferroelectric, $PbZrO_3$", *Advanced Materials* **35**, 2206541 (2023).

[74] Adriana Ladera, Ravi Kashikar, **S. Lisenkov** and I. Ponomareva, "Machine learning reveals memory of the parent phases in ferroelectric relaxors $Ba(Ti_{1-x},Zr_x)O_3$", *Advanced Theory and Simulations* **6**, 2200690 (2023).

2022

[73] Ravi Kashikar, Subhashree Chatterjee, Folarin Shola Taofeek, Abhisikta Barman, Shubhankar Barman, Sohini Kar-Narayan, Anuja Datta, **Sergey Lisenkov**, Devajyoti Mukherjee, and I. Ponomareva, "Hidden variable in the electrocaloric effect of ferroics", *Physical Review Materials* **6**, 124403 (2022).

[72] Partha Sarathi Ghosh, **S. Lisenkov** and I. Ponomareva, "Negative Longitudinal Piezoelectricity Coexisting with both Negative and Positive Transverse Piezoelectricity in Hybrid Formate Perovskite", *ACS Applied Materials and Interfaces* **14**, 46449 (2022).

[71] Ravi Kashikar, P. S. Ghosh, **S. Lisenkov**, A. Stroppa and I. Ponomareva, "Rashba effects in lead-free ferroelectric semiconductor $[CH_3PH_3]SnBr_3$", *Physical Review Materials* **6**, 104603 (2022).

[70] P. S. Ghosh, D. DeTellem, Junyu Ren, S. Witanachchi, Shengqian Ma, **S. Lisenkov**, and I. Ponomareva, "Unusual properties of hydrogen-bonded ferroelectrics: the case of Co-formate", *Physical Review Letters* **128**, 077601 (2022).

2021

[69] Ravi Kashikar, P. S. Ghosh, **S. Lisenkov**, B. R. K. Nanda, and I. Ponomareva, "Chemically and electrically tunable spin-polarizations in ferroelectric Cd-based hybrid organic-inorganic perovskites", *Physical Review B* **104**, 235132 (2021).

[68] P. S. Ghosh, J. Doherty, **S. Lisenkov**, and I. Ponomareva, "Tunability of structure, polarization and band gap of high T_C organic-inorganic ferroelectrics by hydrostatic pressure: first-principles study", *Journal of Physical Chemistry C* **125**, 16296 (2021).

[67] M. Kingsland, P. S. Ghosh, **S. Lisenkov**, and I. Ponomareva, "Structural, electrical, and electromechanical properties of inverse hybrid perovskites from first-principles: The case of $(CH_3NH_3)_3OI$", *Journal of Physical Chemistry C* **125**, 8794 (2021).

2020

[66] **S. Lisenkov**, A. Ladera and I. Ponomareva, "$Ba(Zr_{1-x},Ti_x)O_3$ relaxors: dynamic ferroelectrics at gigahertz", *Physical Review B* **102**, 224109 (2020).

[65] P. S. Ghosh, **S. Lisenkov**, and I. Ponomareva, "Phase switching as the origin of large piezoelectric response in organic-inorganic perovskites: A first-principles study", *Physical Review Letters* **125**, 207601 (2020).

[64] **S. Lisenkov**, Yulian Yao, Nazanin Bassiri-Gharb, and I. Ponomareva, "Prediction of high strain polar phases in antiferroelectric $PbZrO_3$ from multiscale approach", *Physical Review B* **102**, 104101 (2020).

[63] M. Kingsland, K. A. Lynch, **S. Lisenkov**, Xiao He, and I. Ponomareva, "Comparative Study of Minnesota Functionals Performance on Ferroelectric $BaTiO_3$ and $PbTiO_3$", *Physical Review Materials* **4**, 073802 (2020).

2019

[62] C. Mentzer, **S. Lisenkov**, Z. Fthenakis, and I. Ponomareva, "Phase evolution in the ferroelectric relaxor $Ba(Ti,Zr)O_3$ from atomistic simulations", *Physical Review B* **99**, 064111 (2019).

2018

[61] M. Kingsland, **S. Lisenkov** and I. Ponomareva, "Unveiling electrocaloric potential of antiferroelectrics with phase competition", *Advanced Theory and Simulations* **1**, 1800096 (2018).

[60] **S. Lisenkov** and I. Ponomareva, "High-frequency intrinsic dynamics of the electrocaloric effect from direct atomistic simulations", *Physical Review B* **97**, 184104 (2018).

[59] **S. Lisenkov** and I. Ponomareva, "Tuning the electrocaloric effect by varying Sr concentration in ferroelectric $Ba_{1-x}Sr_xTiO_3$", *Physical Review Materials* **2**, 055402 (2018).

2017

[58] P. Jouzdani, S. Cuozzo, **S. Lisenkov**, and I. Ponomareva, "Enhancement of electrocaloric response through quantum effects", *Physical Review B* **96**, 214107 (2017).

[57] R. Herchig, J. Cuozzo, **S. Lisenkov** and I. Ponomareva, "Isentropic magnetoelectric coupling in planar heterostructures", *Applied Physics Letters* **111**, 152904 (2017).

[56] E. Glazkova-Swedberg, J. Cuozzo, **S. Lisenkov**, and I. Ponomareva, "Electrocaloric effect in $PbZrO_3$ thin films with antiferroelectric-ferroelectric phase competition", *Computational Materials Science* **129**, 44 (2017).

2016

[55] **S. Lisenkov**, R. Herchig, Satyanarayan Patel, Rahul Vaish, J. Cuozzo, and I. Ponomareva, "Elastocaloric Effect in Carbon Nanotubes and Graphene", *Nano Letters* **16**, 7008 (2016).

[54] Satyanarayan Patel, Aditya Chauhan, J. Cuozzo, **S. Lisenkov**, I. Ponomareva and Rahul Vaish, "Pyro-paraelectric and flexocaloric effects in barium strontium titanate: A first principles approach", *Applied Physics Letters* **108**, 162901 (2016).

[53] **S. Lisenkov**, B. K. Mani, J. Cuozzo and I. Ponomareva, "Highly tunable piezocaloric effect in antiferroelectric $PbZrO_3$", *Physical Review B* **93**, 064108 (2016).

[52] R. Herchig, B. K. Mani, **S. Lisenkov**, and I. Ponomareva, "Nanoscale properties of $PbZrO_3$ nanowires: phase competition for enhanced energy conversion and storage", *Computational Materials Science* **117**, 468 (2016).

[51] B. K. Mani, R. Herchig, E. Glazkova, **S. Lisenkov**, and I. Ponomareva, "Emergence of ferroelectricity in antiferroelectric nanostructures", *Nanotechnology* **27**, 195705 (2016).

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[49] C.-M. Chang, B. K. Mani, **S. Lisenkov**, and I. Ponomareva, "Prediction of electromagnons in antiferromagnetic ferroelectrics from first-principles: The case of $BiFeO_3$", *Ferroelectrics* **494** (2016).

2015

[48] B. K. Mani, C.-M. Chang, **S. Lisenkov**, and I. Ponomareva, "Critical Thickness for Antiferroelectricity in $PbZrO_3$", *Physical Review Letters* **115**, 097601 (2015).

[47] C.-M. Chang, B. K. Mani, **S. Lisenkov**, and I. Ponomareva, "Thermally Mediated Mechanism to Enhance Magnetoelectric Coupling in Multiferroics", *Physical Review Letters* **114**, 177205 (2015).

[46] B. K. Mani, **S. Lisenkov**, and I. Ponomareva, "Finite-temperature properties of antiferroelectric $PbZrO_3$ from atomistic simulations", *Physical Review B* **91**, 134112 (2015).

[45] E. Glazkova, C.-M. Chang, **S. Lisenkov**, B. K. Mani, and I. Ponomareva, "Depolarizing field in ultrathin electrocalorics", *Physical Review B* **92**, 064101 (2015).

2014

[44] D. Sando, A. Agbelele, C. Daumont, D. Rahmedov, W. Ren, I. C. Infante, **S. Lisenkov**, S. Prosandeev, S. Fusil, E. Jacquet, C. Carretero, S. Petit, M. Cazayous, J. Juraszek, J.-M. Le Breton, L. Bellaiche, B. Dkhil, A. Barthelemy, and M. Bibes, "Control of ferroelectricity and magnetism in multi-ferroic $BiFeO_3$ by epitaxial strain", *Philosophical Transactions of the Royal Society A* **28**, 20120438 (2014).

2013

[43] **S. Lisenkov**, B. K. Mani, C.-M. Chang, J. Almand, and I. Ponomareva, "Multicaloric effect in ferroelectric $PbTiO_3$ from first principles", *Physical Review B* **87**, 224101 (2013).

[42] J. Weerasinghe, L. Bellaiche, T. Ostapchuk, P. Kuzel, C. Kadlec, **S. Lisenkov**, I. Ponomareva, and J. Hlinka, "Emergence of central mode in the paraelectric phase of ferroelectric perovskites", *MRS Communications* **3**, 42 (2013).

[41] Hemant Shah, Austin Carver, Kasun Fernando, Sowmya Kolli, Buddika Abeyweera, **Sergey Lisenkov**, Madhu Menon, and Bruce Alphenaar, "Optical Generation and Detection of Polaronic States in PCBM", *Journal of Physical Chemistry C* **117**, 26538 (2013).

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[39] Antonis N. Andriotis, Giannis Mpourmpakis, **Sergey Lisenkov**, Michael R. Sheetz, and Madhu Menon, "U-calculation of the LSDA+U functional using the hybrid B3LYP and HSE functionals", *Physica Status Solidi B* **250**, 356–363 (2013).

[38] Ekhard K. H. Salje, Michael A. Carpenter, Guillaume F. Nataf, Gunnar Picht, Kyle Webber, Jeevaka Weerasinghe, **S. Lisenkov**, and L. Bellaiche, "Elastic excitations in $BaTiO_3$ single crystals and ceramics: Mobile domain boundaries and polar nanoregions observed by resonant ultrasonic spectroscopy", *Physical Review B* **87**, 014106 (2013).

2012

[37] I. Ponomareva and **S. Lisenkov**, "Bridging the Macroscopic and Atomistic Descriptions of the Electrocaloric Effect", *Physical Review Letters* **108**, 167604 (2012).

[36] I. Ponomareva and **S. Lisenkov**, "Giant elastocaloric effect in ferroelectric $Ba_{0.5}Sr_{0.5}TiO_3$ alloys from first-principles", *Physical Review B* **86**, 104103 (2012).

[35] J. E. Rault, W. Ren, S. Prosandeev, **S. Lisenkov**, D. Sando, S. Fusil, M. Bibes, A. Barthelemy, L. Bellaiche, and N. Barrett, "Thickness-Dependent Polarization of Strained $BiFeO_3$ Films with Constant Tetragonality", *Physical Review Letters* **109**, 267601 (2012).

[34] Antonis N. Andriotis, Ernst Richter, **Sergey Lisenkov**, Michael R. Sheetz, and Madhu Menon, "Electronic Structure, Optical Properties and Electronic Conductivity of Silicon Carbide Nanowires", *Journal of Computational and Theoretical Nanoscience* **9**, 2008 (2012).

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[32] **Sergey Lisenkov**, Antonis N. Andriotis and Madhu Menon, "Magnetic Anisotropy and Engineering of Magnetic Behavior of the Edges in Co Embedded Graphene Nanoribbons", *Physical Review Letters* **108**, 187208 (2012).

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2011

[30] **S. Lisenkov**, I. Ponomareva, and L. Bellaiche, "Central Mode in Disordered $(Ba_{0.6}Sr_{0.4})TiO_3$ Solid Solution from First Principles", *Proceedings of the 2011 DoD High Performance Computing Modernization Program Users Group Conference*, 290 (2011).

[29] R. Michael Sheetz, Ernst Richter, Antonis N. Andriotis, **Sergey Lisenkov**, Chandrashekhar Pendyala, Mahendra K. Sunkara, and Madhu Menon, "Visible-light absorption and large band-gap bowing of $GaN_{1-x}Sb_x$ from first principles", *Physical Review B* **84**, 075304 (2011).

[28] Wei Ren, L. Bellaiche and **S. Lisenkov**, "Tackling Complex Phenomena in Nanoscale Multiferroics", *IEEE Computer Society (Proceedings of the 2010 DoD High Performance Computing Modernization Program Users Group Conference)*, 259 (2011).

[27] **Sergey Lisenkov**, Antonis N. Andriotis, R. Michael Sheetz, and Madhu Menon, "Effects of codoping on the ferromagnetic enhancement in ZnO", *Physical Review B* **83**, 235203 (2011).

[26] Narayani Choudhury, Laura Walizer, **Sergey Lisenkov**, and L. Bellaiche, "Geometric frustration in compositionally modulated ferroelectrics", *Nature* **470**, 513 (2011).

[25] Antonis N. Andriotis, **Sergey Lisenkov** and Madhu Menon, "Ferromagnetic interactions in hosted bipartite materials – generalized-double-exchange and generalized-superexchange interactions", *Journal of Physics: Condensed Matter* **23**, 086004 (2011).

2010

[24] T. Ostapchuk, J. Petzelt, P. Kuzel, M. Savinov, J. Hlinka, A. Tkach, P. Vilarinho, **Sergey Lisenkov**, I. Ponomareva and L. Bellaiche, "Lattice dynamics in $Ba_{0.7}Sr_{0.3}TiO_3$: study by THz and IR spectroscopy and ab initio simulations", *Phase Transitions* **83**, 955 (2010).

[23] **Sergey Lisenkov**, Antonis N. Andriotis and Madhu Menon, "Magnetic graphene: A new class of cages formed from graphene sheets and carbon nanotubes", *Physical Review B* **82**, 165454 (2010).

[22] S. Prosandeev, **S. Lisenkov**, and L. Bellaiche, "Kittel Law in $BiFeO_3$ Ultrathin Films: A First-Principles-Based Study", *Physical Review Letters* **105**, 147603 (2010).

[21] I. C. Infante, **S. Lisenkov**, B. Dupe, M. Bibes, S. Fusil, E. Jacquet, G. Geneste, S. Petit, A. Courtial, J. Juraszek, L. Bellaiche, A. Barthelemy, and B. Dkhil, "Bridging Multiferroic Phase Transitions by Epitaxial Strain in $BiFeO_3$", *Physical Review Letters* **105**, 057601 (2010).

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[19] D. Albrecht, **S. Lisenkov**, Wei Ren, D. Rahmedov, Igor A. Kornev and L. Bellaiche, "Ferromagnetism in multiferroic $BiFeO_3$ films: A first-principles-based study", *Physical Review B (Rapid Communication)* **81**, 140401(R) (2010).

2009

[18] **S. Lisenkov** and I. Ponomareva, "Intrinsic electrocaloric effect in ferroelectric alloys from atomistic simulations", *Physical Review B* **80**, 140102(R) (2009).

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[16] T. Ostapchuk, J. Petzelt, J. Hlinka, V. Bovtun, P. Kuzel, I. Ponomareva, **S. Lisenkov**, L. Bellaiche, A. Tkach and P. Vilarinho, "Broad-band dielectric spectroscopy and ferroelectric soft-mode response in the $Ba_{0.6}Sr_{0.4}TiO_3$ solid solution", *Journal of Physics: Condensed Matter* **21**, 474215 (2009).

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[14] **S. Lisenkov**, D. Rahmedov and L. Bellaiche, "Electric-Field-Induced Paths in Multiferroic $BiFeO_3$ from Atomistic Simulations", *Physical Review Letters* **103**, 047204 (2009).

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2008

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2007

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2006

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